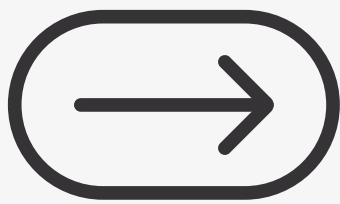




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The Vector DB Playbook



A quick way to choose the right vector store for your AI workloads.

Vector dbs covered

01 Pinecone

02 Weaviate

03 Milvus

04 Qdrant

05 Chroma

06 Faiss



What is a Vector Database?

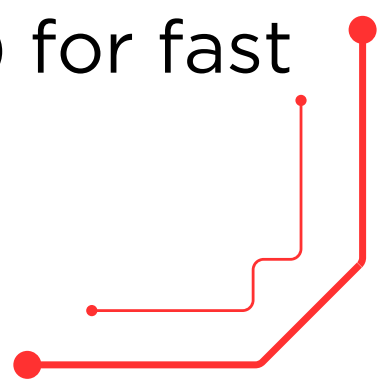
A vector database stores embeddings and helps you search by similarity instead of exact matches. Instead of looking for exact text or values, it finds the closest meaning using vector distances.

How it differs from normal databases

- Normal DBs work with structured rows and columns
- They retrieve exact matches (SQL queries, filters)
- Vector DBs work with embeddings (floating-point vectors)
- They retrieve similar items using distance metrics
- They use indexing like HNSW, IVF, PQ to speed up similarity search

Best techniques to follow

- Pick the right index (HNSW for accuracy, IVF/PQ for speed)
- Normalize vectors when using cosine similarity
- Use metadata filtering to reduce candidate set
- Choose proper chunk size before embedding (256–512 tokens)
- Run periodic index rebuilds for large datasets
- Use quantization only when memory is limited
- Use ANN (Approximate Nearest Neighbor) for fast queries



1. Pinecone

Fully managed, cloud-native vector database designed for production-scale similarity search.

Indexing Techniques

- Proprietary indexing with automatic optimization
- Pod-based architecture for horizontal scaling
- Supports metadata filtering with pre-filtering

Similarity Search Methods

- Cosine similarity
- Euclidean distance (L2)
- Dot product

Key Features

- Real-time updates without rebuild
- Vector + metadata filtering
- Built-in backup and disaster recovery

Best For

- Production applications requiring high availability
- Teams wanting minimal infrastructure management
- Startups to enterprise scale

2. Weaviate

Open-source vector database with GraphQL API and built-in vectorization modules.

Indexing Techniques

- HNSW (Hierarchical Navigable Small World) graphs
- Dynamic index updates
- Configurable HNSW parameters (efConstruction, maxConnections)

Similarity Search Methods

- Cosine similarity
- Euclidean distance
- Dot product
- Manhattan distance

Key Features

- Multi-modal search (text, images, etc.)
- Built-in vectorization modules (OpenAI, Cohere, HuggingFace)
- Semantic search with keyword boosting

Best For

- Applications requiring multi-modal search
- Projects needing semantic understanding out-of-the-box
- Hybrid search scenarios (keyword + vector)

3. Milvus

Open-source vector database built for scalability and performance, part of the LF AI & Data Foundation.

Indexing Techniques

- Multiple index types: HNSW, IVF_FLAT, IVF_SQ8, IVF_PQ, RHNSW_FLAT
- GPU-accelerated indexing options
- Dynamic segment-based architecture

Similarity Search Methods

- Cosine similarity
- Euclidean distance (L2)
- Inner product (IP)
- Jaccard distance
- Hamming distance

Key Features

- Horizontal scalability with distributed architecture
- Support for billions of vectors
- GPU acceleration

Best For

- Large-scale applications (billions of vectors)
- High-throughput search requirements
- Applications requiring GPU acceleration

4. Qdrant

Open-source vector database written in Rust, emphasizing performance and developer experience.

Indexing Techniques

- HNSW with custom optimizations
- Payload indexing for efficient filtering
- Quantization support (scalar, product quantization)

Similarity Search Methods

- Cosine similarity
- Euclidean distance
- Dot product

Key Features

- Rich filtering capabilities with payload
- Full CRUD operations with snapshots
- Recommend endpoint for similar item discovery

Best For

- Applications requiring complex filtering
- Production systems needing reliability
- Projects with memory constraints (quantization)

5. Chroma

Open-source embedding store designed for AI applications, emphasizing simplicity and developer experience.

Indexing Techniques

- HNSW (via hnswlib)
- In-memory and persistent storage options
- Lightweight embedding store (not a full-featured distributed vector DB)

Similarity Search Methods

- Cosine similarity
- L2 (Euclidean distance)
- Inner product

Key Features

- Lightweight and easy to set up
- Built-in embedding function integrations
- Local-first design for development

Best For

- Prototyping and development
- Small to medium-scale applications
- AI/ML projects and notebooks

6. Faiss

(Facebook AI Similarity Search)

Library by Meta for efficient similarity search, not a full database but widely used as a foundation.

Indexing Techniques

- Extensive index types: Flat, IVF, HNSW, PQ (Product Quantization), LSH
- GPU-accelerated indexes
- Composite indexes (e.g., IVF + PQ)

Similarity Search Methods

- L2 (Euclidean distance)
- Inner product
- Cosine similarity

Key Features

- Extremely fast search performance
- GPU acceleration support
- Memory-efficient quantization

Best For

- Maximum performance requirements
- Research and experimentation
- Custom solutions (building your own DB)

Choosing the Right Vector DB

Choose Pinecone if:

- You want fully managed, zero-ops solution
- Budget allows for managed services
- Need enterprise SLAs and support

Choose Weaviate if:

- Multi-modal search is important
- You need semantic understanding out-of-the-box
- GraphQL API fits your architecture

Choose Milvus if:

- Scale is your primary concern (billions of vectors)
- You need GPU acceleration
- You have DevOps resources for management

Choose Qdrant if:

- Complex filtering is essential
- You want balance of features and performance
- Rust-based reliability appeals to you

Choose Chroma if:

- Rapid prototyping is the goal
- Simple integration with AI frameworks
- Working on smaller-scale projects

Choose Faiss if:

- Building custom solutions
- Maximum performance is critical
- You have ML engineering expertise

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